Spencer Hill

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Education

PhD in Computer Science, University of Toronto 2026 – Present

- Supervisors: Chris Maddison and Murat Erdogdu (Funding: Ontario Graduate Scholarship)

MASc in Mathematics and Engineering, Queen's University 2024 – 2026

- Supervisors: Fady Alajaji and Tamás Linder (Funding: NSERC CGRS and Ontario Graduate Scholarship)
- Thesis: Channel Simulation: Fundamental Limits and Optimal Algorithms

BASc in Mathematics and Computer Engineering, Queen's University 2019 – 2023

- GPA: 4.27/4.3 (2nd in Class of 800, University Medal in Mathematics and Engineering)
- Thesis: Data Compression via Nonlinear Transform Coding using Artificial Neural Networks

Research Experience and Projects

Master's Thesis, Queen's University (Supervisors: Tamás Linder and Fady Alajaji) Sept. 2024 – Aug. 2026

- Researching channel simulation algorithms for use in neural-network-based learned image compression.
- Generalized channel simulation to an exponential communication cost and designed a $\mathcal{O}(N \log N)$ channel simulation algorithm for discrete channels; the previous best complexity was exponential in N.

Reinforcement Learning Researcher, Natural Resources Canada Nov. 2024 – Present

- Theoretical and experimental investigation into reinforcement learning for optimal surveillance of invasive species. I am the principal investigator, collaborating with Jue Wang and Denys Yemshanov.
- Developed an optimal algorithm for allocating surveillance resources under partial information; this work was presented at the 2025 INFORMS Annual Meeting and is in preparation to be submitted to Management Science.

Visiting Researcher, Pompeu Fabra University (Host: Gergely Neu) Oct. 2025 - Nov. 2025

- Researched adaptive data analysis through the lens of PAC-Bayes generalization bounds.
- Showed classical results could be recovered under this new framework, and proved the generalization of adaptive learning without distortion for smooth loss functions.

Undergraduate Thesis, Queen's University (Supervisor: Tamás Linder) Sept. 2022 – Apr. 2023

- Mathematically formulated and implemented (in Python) a nonlinear transform coding system that uses artificial neural networks to compress CT scan images.
- Awarded the Keyser Prize for outstanding thesis among Mathematics and Engineering students.

Quantum Machine Learning Research Project Lead, QMIND, Queen's AI Hub May 2021 – Apr. 2022

- Led a team of undergraduate and graduate students to research image generation with Quantum Generative Adversarial Networks on a simulated Quantum Computer.
- Improved state-of-the-art methods by applying novel data encoding techniques that maintained quantitative performance while dramatically improving qualitative results.
- Presented work at CUCAI 2022 and published in the conference proceedings.

Machine Learning Researcher, Queen's University (Supervisor: Tucker Carrington) May 2021 – Aug. 2021

- Created a local Gaussian Processes method that reduced the regression complexity from $\mathcal{O}(N^3)$ to $\mathcal{O}(m^3 + N)$ (where $m \ll N$), yielding a 1,000x improvement in computational efficiency.
- Applied our novel method to solve previously intractable problems in computational chemistry and published in the Journal of Molecular Spectroscopy.

Machine Learning Researcher, QMIND, Queen's AI Hub Sept. 2020 –Apr. 2021

- Designed and implemented a tree-based graph convolutional generative adversarial network capable of generating 3D point clouds, which was applied to a novel domain in medical imaging (3D body scans).
- Presented work at CUCAI 2021 and published in the conference proceedings.

Publications

(Journal Articles)

S. Hill, F. Alajaji, and T. Linder, "Communication complexity of exact sampling under Rényi information," *IEEE Transactions on Information Theory*, vol. 72, no. 6, pp. 3525-3538, June 2026.

N. Yang, **S. Hill**, S. Manzhos, and T. Carrington, "A local Gaussian Processes method for fitting potential surfaces that obviates the need to invert large matrices," *Journal of Molecular Spectroscopy*, vol. 393, pp. 111774–111774, Mar. 2023.

(Conference Papers)

S. Hill, F. Alajaji, T. Linder, and G. Flamich, "Rejection Sampling is Optimal for Relative Entropy Coding," *Information Theory Workshop,* 2026.

G. Flamich and **S. Hill**, "Singular Relative Entropy Coding with Bits Back Rejection Sampling," **Spotlight Presentation** at *Learn to Compress and Compress to Learn Workshop at International Symposium on Information Theory*, 2026.

S. Hill, F. Alajaji, and T. Linder, "Exact Channel Simulation under Exponential Cost," *Proc. 61st Allerton Conference on Communication, Control, and Computing*, 2025.

S. Hill, A. Grace, J. Ewaniuk, C. Estabrooks, and J. Otten, "Improved Encoding Methods for Quantum Generative Adversarial Networks", *Proc. Canadian Undergraduate Conference on AI*, 2022.

E. Bonnardeaux, **S. Hill**, D. Stewart, and N. Cabral, "3D Human Body Shape Generation," *Proc. Canadian Undergraduate Conference on AI*, 2021.

(In Preparation)

S. Hill, J. Wang, and D. Yemshanov, "Catch Me If You Can: Monitoring A Cryptic Spreading Invader."

S. Hill, F. Aljaji, and T. Linder, "Fast Channel Simulation for Arbitrary Discrete Channels."

S. Hill, G. Flamich, F. Alajaji, and T. Linder, "Functional Information, Optimal Channel Simulation and Lossy Source Coding."

Conference Presentations

"Rejection Sampling is Optimal for Relative Entropy Coding," Hungarian Machine Learning Days, Budapest, Hungary, 2026.

"Catch Me if You Can: An Optimal Pest Survey Strategy to Delimit Biological Invasions," Canadian Operational Research Society Annual Meeting, Kingston, Canada, 2026.

"Communication Complexity of Exact Sampling under Exponential Cost," Canadian Mathematical Society Winter Meeting: Mathematics of Machine Learning Session, Toronto, Canada, 2025.

"Catch Me If You Can: An Optimal Pest Survey Strategy to Delimit Biological Invasions," INFORMS Annual Meeting, Atlanta, Georgia, 2025.

"Exact Channel Simulation under Exponential Cost," 61st Allerton Conference on Communication, Control, and Computing, Urbana-Champaign, United States, 2025.

"Communication Complexity of Exact Sampling under Rényi Information," Hungarian Machine Learning Days, Budapest, Hungary, 2025.

"Exact Channel Simulation under Exponential Cost," Poster in the Recent Results Session at the International Symposium on Information Theory (ISIT), Ann Arbor, United States, 2025.

"Nonlinear Transform Coding using Artificial Neural Networks," Canadian Undergraduate Conference on AI, Kingston, Canada, 2023.

"Quantum Generative Adversarial Networks," Canadian Undergraduate Conference on AI, Virtual, 2022.

"Local Gaussian Process Regression for Interatomic Potentials," Materials Research Society Fall Meeting and Exhibit, Boston, United States, 2021.

"3D Human Body Shape Generation," Canadian Undergraduate Conference on AI, Virtual, 2021.

Select Awards and Scholarships

Ontario Graduate Scholarship ($15,000), 2026 – 2027. Host Institution: University of Toronto.

University of Toronto Faculty of Arts and Sciences (FAST) Doctoral Award, ($20,000), 2026 – 2030.

Ontario Graduate Scholarship ($15,000), 2025 – 2026. Host Institution: Queen's University.

NSERC Canada Graduate Scholarship – Master's ($27,000), 2024 – 2025. Host Institution: Queen's University.

Queen's University Medal in Mathematics and Engineering, June 2023.

Keyser Project Prize ($1000), Spring 2023. *Outstanding performance in the final year thesis project.*

Ontario Professional Engineers Foundation Scholarship ($1500), 2022. *Academic excellence and demonstrated leadership of an Ontario engineering student.*

Frank B. Lee Memorial Scholarship in Engineering ($13,000), 2021. *Academic achievement, creative and original thinking and proven leadership skills (1 of 800).*

Bronze Medal at the Canadian Engineering Competition, January 2021. *Placed 3rd in the Junior Design category at the 2021 Canadian Engineering Competition.*

Andrew McMahon Standards of Excellence Award ($3600), Fall 2020. *Academic achievement, strong interpersonal skills, and a commitment to excel in all aspects of university life.*

Royal Conservatory of Music Grade Eight Piano Certificate, 2015.

Teaching Experience

Information Theory (4th Year/Graduate Course), Led tutorials	Fall 2024 and Fall 2025
Probability (3rd Year), Led tutorials	Fall 2025
Mathematics of Engineering Systems (3rd Year), Led tutorials	Winter 2025
Control Theory (4th Year/Graduate Course), Led lab sessions	Fall 2024
Linear Algebra (1st Year), Led tutorials	Winter 2023
Engineering Design (1st Year), Supervised 8 machine learning project teams	Fall 2021 and Winter 2022

Professional Experience

Senior Associate Consultant, Stroud International — Sept. 2023 – Aug. 2024

- Led client teams to drive rapid, significant, and lasting improvements in manufacturing processes. Select client results include:
  - Saved $1.1 million in annual labour cost for a personal care production facility through a 10% improvement in plant-wide UPLH in 6 weeks.
  - Increased throughput of an aerosol line by 50% in 12 weeks with no capital expenditure, worth $600K in annual efficiency savings.
  - Reduced compounding waste in a production facility by 45% in 8 weeks, worth $480K in annual raw material savings.

Managing Director of Design, QMIND, Queen's AI Hub — May 2022 – Apr. 2023

- President of Canada's largest undergraduate machine learning organization, consisting of over 250 students and 35 machine learning projects (including 10 client-facing pro bono consulting projects).
- Organized a national AI conference with 300+ attendees and was chief editor of the conference proceedings.

Technical Skills

Languages: Python, MATLAB, LaTeX, C, C++, Java, SQL.

Frameworks and Tools: Git, TensorFlow, PyTorch, SciPy.